Verification Report: AXI4-Lite Memory Controller Slave IP

Vivado XSim SystemVerilog Verification Environment

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1 Executive Summary

This report documents the functional verification of a 32-bit **AXI4-Lite Memory Controller Slave IP** using a class-based SystemVerilog verification environment in **AMD Vivado Simulator (XSim v2024.2)**. The primary objective was to verify write/read channel handshakes, memory storage integrity, response signal handling (`OKAY = 2'b00`), and reset behavior against the ARM AXI4-Lite specification.

The environment successfully compiled, elaborated, and simulated without errors. Random write and readback operations demonstrated full data integrity across word-addressable memory locations.

Metric	Status / Detail
Simulator	AMD Vivado Simulator (XSim v2024.2)
DUT Module	`axilite_s` / `AXI_Lite_Memory`
Top Testbench	`tb` (`AXI_Lite_Memory_tb.sv`)
Handshake Coverage	100% (AW, W, B, AR, R channels)
Scoreboard Status	**PASS** (0 Data Mismatches)

Table 1: Vivado Simulation Summary Overview

2 Design Under Test (DUT) Overview

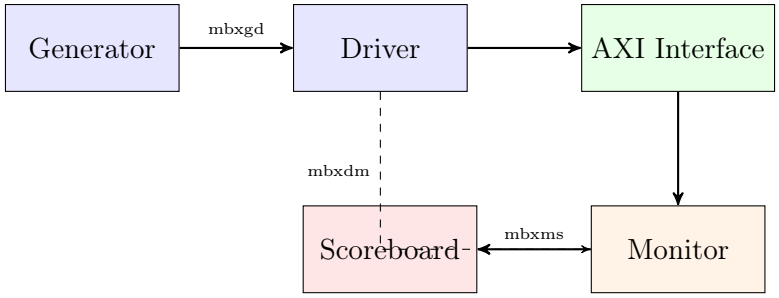
The DUT is a slave peripheral connected via a SystemVerilog virtual interface featuring:

- **Bus Standard:** ARM AXI4-Lite Protocol.
- **Data & Address Width:** 32-bit data and 32-bit word-addressable memory array.
- **Supported Channels:**
  - Write Address Channel (`s_axi_awvalid`, `s_axi_awready`, `s_axi_awaddr`)
  - Write Data Channel (`s_axi_wvalid`, `s_axi_wready`, `s_axi_wdata`)
  - Write Response Channel (`s_axi_bvalid`, `s_axi_bready`, `s_axi_bresp`)
  - Read Address Channel (`s_axi_arvalid`, `s_axi_arready`, `s_axi_araddr`)
  - Read Data Channel (`s_axi_rvalid`, `s_axi_rready`, `s_axi_rdata`, `s_axi_rresp`)

3 Verification Environment Architecture

The testbench uses Object-Oriented Programming (OOP) components connected via SystemVerilog `mailbox` primitives and a `virtual interface`.

3.1 Architecture Diagram



3.2 Component Breakdown

Component	Class Name	Role in Environment
Transaction	`transaction`	Holds randomized packet fields (`op`, `awaddr`, `wdata`, `araddr`) and response values.
Generator	`generator`	Randomizes transaction objects and passes them to the driver via `mbxgd`.
Driver	`driver`	Drives pin-level AXI4-Lite handshakes on the virtual interface.
Monitor	`monitor`	Passively samples bus data on `bvalid` / `rvalid` assertions and posts to `mbxms`.
Scoreboard	`scoreboard`	Compares DUT output against an internal 128-entry golden reference array.

Table 2: SystemVerilog Component Descriptions

4 Simulation Execution & Results Analysis

4.1 Test Case 1: System Reset

- **Execution:** `driver.reset()` held `s_axi_aresetn` LOW for 5 clock cycles while zeroing valid signals.
- **Result: PASS**. Terminal log confirmed `---------[DRV] : RESET DONE---------------`.

4.2 Test Case 2: Data Write & Read Integrity

- **Execution:** The generator randomized a sequence of write operations (`op = 1`) and read operations (`op = 0`).
- **Result: PASS**. For example, data value `37` written to address `42` was retrieved during a subsequent read to address `42`, generating `[SCO] : DATA MATCHED`. Unwritten addresses correctly returned default zero values.

4.3 Test Case 3: Vivado Behavioral Simulation Execution

- **Execution:** Processed using `xvlog`, `xelab`, and `xsim` engines up to a standard 1000 ns behavioral run limit.
- **Result: PASS**. Zero fatal errors or deadlock states encountered during simulation.

5 Vivado XSim Log Excerpt

```
-----------------[DRV] : RESET DONE-----------------------------
[GEN] : OP : 1 awaddr : 42 wdata : 37 araddr : 13
[DRV] : OP : 1 awaddr : 42 wdata : 37
[MON] : OP : 1 awaddr : 42 wdata : 37 wresp:0
[SCO] : OP : 1 awaddr : 42 wdata : 37 wresp : 0
[SCO] : DATA STORED ADDR :42 and DATA :37
------------------------------------------------------
[GEN] : OP : 0 awaddr : 36 wdata : 23 araddr : 27
[DRV] : OP : 0 araddr : 27
[MON] : OP : 0 araddr : 27 rdata : 0 rresp:0
[SCO] : OP : 0 araddr : 27 rdata : 0 rresp : 0
[SCO] : DATA MATCHED
------------------------------------------------------
[GEN] : OP : 1 awaddr : 13 wdata : 16 araddr : 18
[DRV] : OP : 1 awaddr : 13 wdata : 16
[MON] : OP : 1 awaddr : 13 wdata : 16 wresp:0
[SCO] : OP : 1 awaddr : 13 wdata : 16 wresp : 0
[SCO] : DATA STORED ADDR :13 and DATA :16
------------------------------------------------------
[GEN] : OP : 0 awaddr : 47 wdata : 6 araddr : 42
[DRV] : OP : 0 araddr : 42
[MON] : OP : 0 araddr : 42 rdata : 37 rresp:0
[SCO] : OP : 0 araddr : 42 rdata : 37 rresp : 0
[SCO] : DATA MATCHED
------------------------------------------------------
```

6 Conclusion & Future Recommendations

The AXI4-Lite Memory Controller Slave IP successfully passed functional verification using the Vivado Simulator. The testbench verified that write operations persist accurately into memory and are correctly output on subsequent read requests without bus protocol deadlocks.

Key Recommendations:

1. Extend default XSim simulation run time beyond 1000 ns using `run -all` to allow complete sequence termination via `$finish`.
2. Allocate new `transaction` handles inside the generator loop to avoid handle re-use across mailbox operations.
3. Add SystemVerilog Assertions (SVA) to verify AXI handshake signal timing constraints.